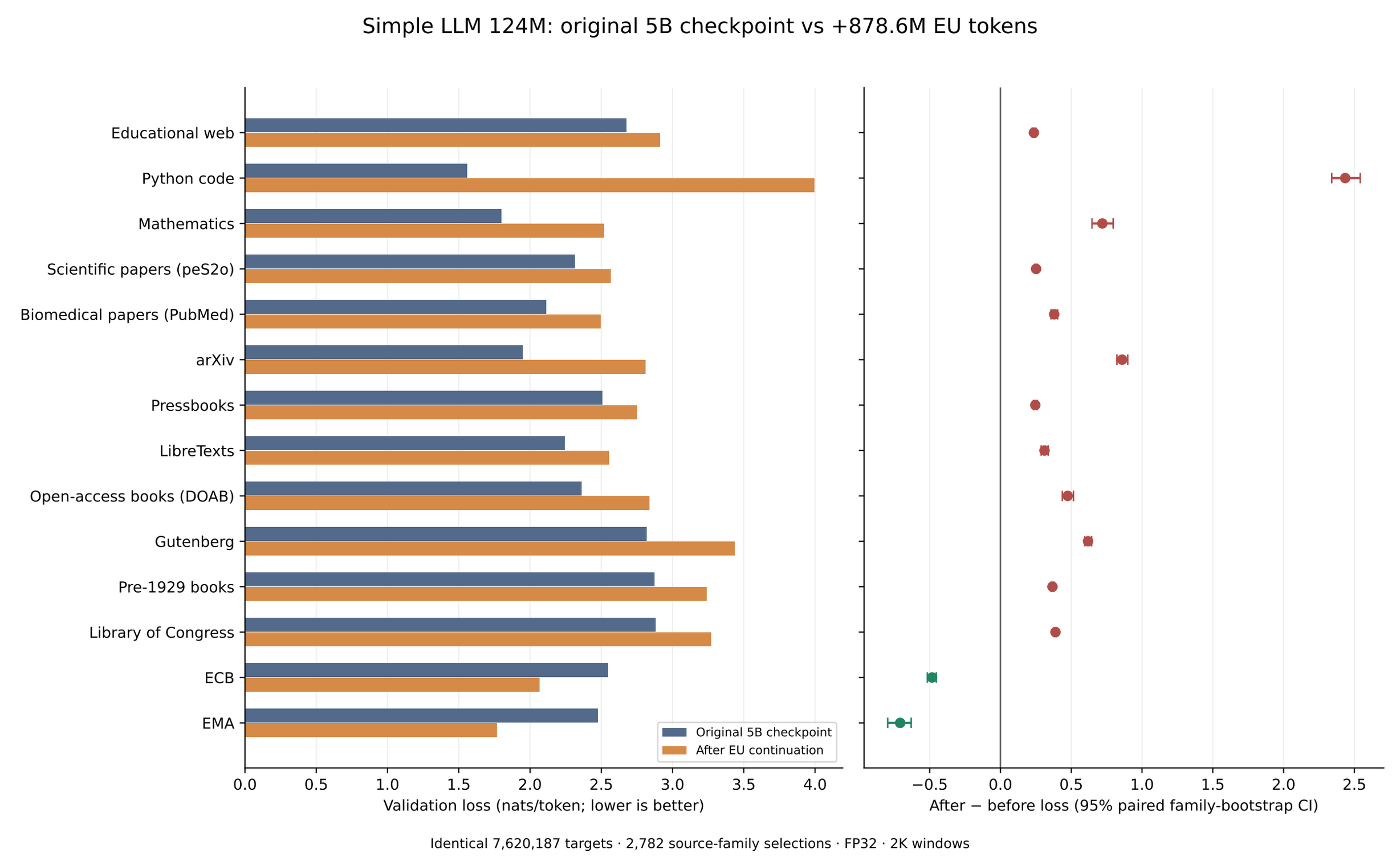
Simple LLM 124M: original 5B checkpoint vs +878.6M EU tokens
Educational web
Python code
Mathematics
Scientific papers (peS2o)
Biomedical papers (PubMed)
arXiv
Pressbooks
LibreTexts
Open-access books (DOAB)
Gutenberg
Pre-1929 books
Library of Congress
ECB
EMA
Original 5B checkpoint
After EU continuation
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Validation loss (nats/token; lower is better)
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
After − before loss (95% paired family-bootstrap CI)
Identical 7,620,187 targets · 2,782 source-family selections · FP32 · 2K windows